ClipSAT : Math for AI & Data Science

Unit 6.01: Linear Regression & OLS

Name: _____

Q1. A fitted simple linear regression model is hat{y} = 7x + 18 . What is the predicted target value hat{y} when the input feature is x = 4 ?

[ ] A: 46
[ ] B: 49
[ ] C: 42
[ ] D: 28

Q2. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 20 and the Total Sum of Squares is text{SST} = 100 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Q3. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 40 and the Total Sum of Squares is text{SST} = 200 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Q4. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 60 and the Total Sum of Squares is text{SST} = 300 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Q5. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 80 and the Total Sum of Squares is text{SST} = 400 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

Q6. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 100 and the Total Sum of Squares is text{SST} = 500 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80
[ ] B: R^2 = 0.20
[ ] C: R^2 = 0.90
[ ] D: R^2 = 0.25

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Q7. In a multiple linear regression model, the Sum of Squared Errors is text{SSE} = 120 and the Total Sum of Squares is text{SST} = 600 . What is the coefficient of determination R^2 ?

[ ] A: R^2 = 0.80

[ ] B: R^2 = 0.20

[ ] C: R^2 = 0.90

[ ] D: R^2 = 0.25

Q8. For a bivariate feature dataset, text{Cov}x, y = 12 , text{Var}x = 4 , bar{x} = 5 , and bar{y} = 25 . Compute the OLS regression slope hat{beta}_1 and intercept hat{beta}_0 .

[ ] A: hat{beta}_1 = 0.3333333333333333, ;

[ ] B: hat{beta}_1 = 3, ; hat{beta}_0 = 10

[ ] C: hat{beta}_1 = 3, ; hat{beta}_0 = 25

[ ] D: hat{beta}_1 = 4, ; hat{beta}_0 = 5